is chosen may be historical accident, or it may be a matter of more deliberate planning by chemists. In nature's chemistry the choice will never, of course, be a deliberate one. Instead it will come about through natural selection. But how can natural selection see to it that the two pathways are not mixed, and that cooperating groups of compatible genes emerge? In very much the same way as I suggested with my analogy of the German and English rowers (Chapter 5). The important thing is that a gene for a stage in pathway 1 will flourish in the presence of genes for other stages in pathway 1, but not in the presence of pathway 2 genes. If the population already happens to be dominated by genes for pathway 1, selection will favour other genes for pathway 1, and penalize genes for pathway 2. And vice versa. Tempting as it is, it is positively wrong to speak of the genes for the six enzymes of pathway 2 being selected 'as a group'. Each one is selected as a separate selfish gene, but it flourishes only in the presence of the right set of other genes.

Nowadays this cooperation between genes goes on within cells. It must have started as rudimentary cooperation between self-replicating molecules in the primeval soup (or whatever primeval medium there was). Cell walls perhaps arose as a device to keep useful chemicals together and stop them leaking away. Many of the chemical reactions in the cell actually go on in the fabric of membranes; a membrane acts as a combined conveyor belt and test-tube rack. But cooperation between genes did not stay limited to cellular biochemistry. Cells came together (or failed to separate after cell division) to form many-celled bodies.

This brings us to the second of my three questions. Why did cells gang together; why the lumbering robots? This is another question about cooperation. But the domain has shifted from the world of molecules to a larger scale. Many-celled bodies outgrow the microscope. They can even become elephants or whales. Being big is not necessarily a good thing: most organisms are bacteria and very few are elephants. But when the ways of making a living that are open to small organisms have all been filled, there are still prosperous livings to be made by larger organisms. Large organisms can eat smaller ones, for instance, and can avoid being eaten by them.

The advantages of being in a club of cells don't stop with size. The cells in the club can specialize, each thereby becoming more efficient at performing its particular task. Specialist cells serve other cells in the club and they also benefit from the efficiency of other specialists. If there are